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Roll No

EX-8004 (1) (CBGS)

B.E. VIII Semester Examination, June 2020

Choice Based Grading System (CBGS)

VLSI Circuits and Systems

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions out of eight questions.

ii) All questions carry equal marks.

iii) Assume suitable data, if required.

1. a) Give a brief introduction on CMOS VLSI circuits. Give some practical applications of VLSI circuits.
b) Draw and explain some differences between the N switches and P switches.
2. a) Draw and explain the basic block diagram of VLSI design flow.
b) Explain CMOS design strategies with examples.
3. a) Write some specifications of sequential systems. Explain any characterizing equation of VLSI circuit.
b) Write state table of any MOS Circuit and explain its transition diagram.
4. a) Define synchronous sequential machines. Write down some of its applications.
b) Draw and explain Moore model using any of the CMOS structure.
5. a) Explain the races and hazards problem in sequential machines.
b) Explain the minimization of the state table of completely specified sequential machines.
6. a) Write down some differences between the fundamental mode and pulse mode asynchronous sequential machines.
b) Explain the concept of secondary state assignments in asynchronous sequential machines.
7. a) What do you mean by faults in CMOS circuits? Explain in brief.
b) Write down the steps to detect fault using Boolean difference method in CMOS circuit.
8. Write short notes on any four:
 - a) Regularity
 - b) Modularity
 - c) Introduction to CAD tool
 - d) PALASM
 - e) CPLD
